

Cybercriminals leveraging superheroes

A McAfee study reveals that cybercriminals are using superheroes searches to unleash malware – Aquaman found the most toxic.



12 rules of Internet

While living on the Web, these are some rules (unofficial) which are really worth considering. <http://dgit.in/145LhJi>

NOKIA LUMIA 1020:

Tech behind the 41 MP PureView Camera Explained

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Mid-July, Stephen Elop took the stage to announce to the world the Nokia Lumia 1020, a Windows Phone 8 device with a camera like no other. Nokia had already established itself as the big dog of cellphone imaging with the 41 megapixel 808 PureView camera they launched back in February of last year.

We've already known almost everything about the Lumia 1020, thanks to the number of leaks (and pretty major ones at that!) and Elop's announcement proved that almost all of the leaks were spot-on. It has the 41 megapixel sensor, the 6-element Zeiss lens with OIS and an aperture of f/2.2. It also has 2 GB of RAM and 32GB of on-board storage, with no option for expansion. The phone also manages to maintain a rather slim profile, keeping its size just about the same as the Lumia 920, with only a very slight bulge towards the top to house the camera module, which happens to be the real highlight of the phone.

Nokia's managed to keep the size and weight of the phone at the lower end of the weight spectrum – certainly an impressive feat. If one was to look at the PureView



Behold the new Nokia Lumia 1020 and its bling

808 and the Lumia 1020 side-by-side, you'd see the difference. But at the heart of this Windows Phone 8 powered device, is a testament of how engineering can overcome the impossible. So if you're wondering what it really is that Nokia has achieved with the Lumia 1020, let us take you on a guided tour.

The sensor

The Lumia 1020 houses a large 1/1.5" sensor, which is comparatively just a touch smaller than the 1/1.2" silicon chip found in the PureView 808. That does mean that it has relatively smaller pixels (1.1 microns on the Lumia 1080 vs. 1.4 microns on the PureView 808), but what Nokia's done to better this new sensor is employ the BSI design. Since Nokia has established itself as the pinnacle in low-light performance, it was only a matter of time they went the BSI sensor route. The large sensor combined with the BackSide-Illuminated design will definitely help the Lumia 1020 push the boundaries of low light performance.

The optics

The Lumia 1020 houses a lens made of 6 Zeiss floating elements. Most conventional camera phones use a single element inside their camera assemblies to cut costs, but the Lumia 1020 reaches out to the best lens

maker in the world to design something that would complement the quality of their large sensor. The Lumia 1020's 6 element lens is made of 5 high performance plastic elements and one high precision glass element to a barrel that optimizes the way light falls on the sensor. The optical assembly also incorporates optical image stabilization, which we saw first in the Lumia 920. However, the OIS mechanism has also been developed further, with Nokia ditching the previous model in favour of one that now utilizes a barrel shift actuator (using ball bearings) that is designed specifically to move complex optical assemblies with high speed and precision.

The resolution saga

The 41-megapixel sensor on the Lumia 1020 will give you a 5 megapixel image when you shoot using the stock camera app. However, the 5 megapixels you get aren't generated using only that many pixels. Nokia utilizes an oversampling method which allows every pixel to collect data from seven pixels around it, to generate the output of that one particular pixel. What that means is each pixel from the Lumia 1020 is delivering detail and clarity that you would normally get from a photosite comprised of 7 pixels. The image is cap-



Elop showing off the new Lumia 1020



The phone's USP is undoubtedly its camera

tured using every single pixel on the sensor (effectively 41.3MP in 4:3 and 33.8MP in 16:9), and the pixels are extrapolated into a 5 megapixel image to yield detail like you've never seen before.